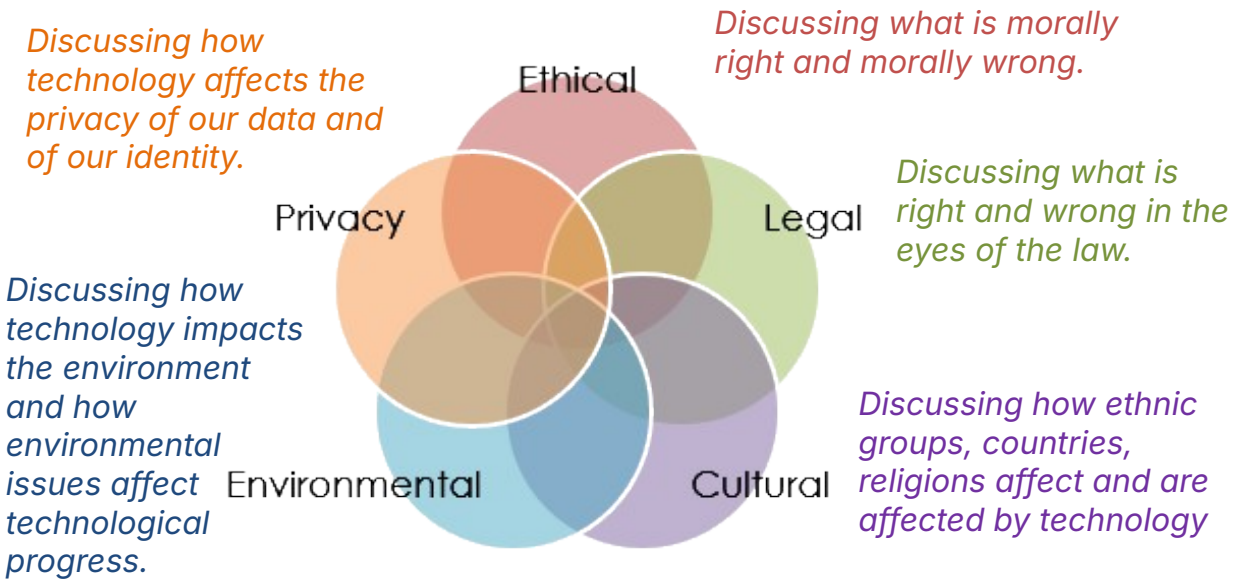
Issues and Legislation

Issues Surrounding Technology

There are a number of different issues surrounding technology and these issues can be categorised as ethical, legal, privacy, environmental and cultural. Over the next few pages we will explore examples of a range of issues, but it must be noted that most issues involving technology encompass a mixture of these categories.



Stakeholders

As we take a look at a number of different technology related issues, it is important that we view these issues through the eyes of the various stakeholders involved. A stakeholder is a person or group who is affected by an issue. For example, when discussing issues relating to illegally downloading music, there will be several stakeholders involved:

- The Individual (downloading the music): they will breaking the law and you could argue that they are being unethical.
- The Music Company: they will lose out on revenue
- The Government: they will be expected to prevent the ability to download illegally but this will be difficult if the site is hosted outside their authority.

You will be expected to discuss a variety of issues in your exam and your discussions need to be from all sides – so always identify the stakeholders and work out how they are affected by the issue.

Technology and the Environment

Obviously electrical technology requires electricity.

This is fine as long as the methods of electricity production are not harmful but unfortunately most electricity is created from the burning of fossil fuels.

Furthermore, the manufacture of components produces waste products which can adversely affect the environment. Technology develops rapidly which means that after a few years, devices become obsolete and are taken to landfill, further causing environmental

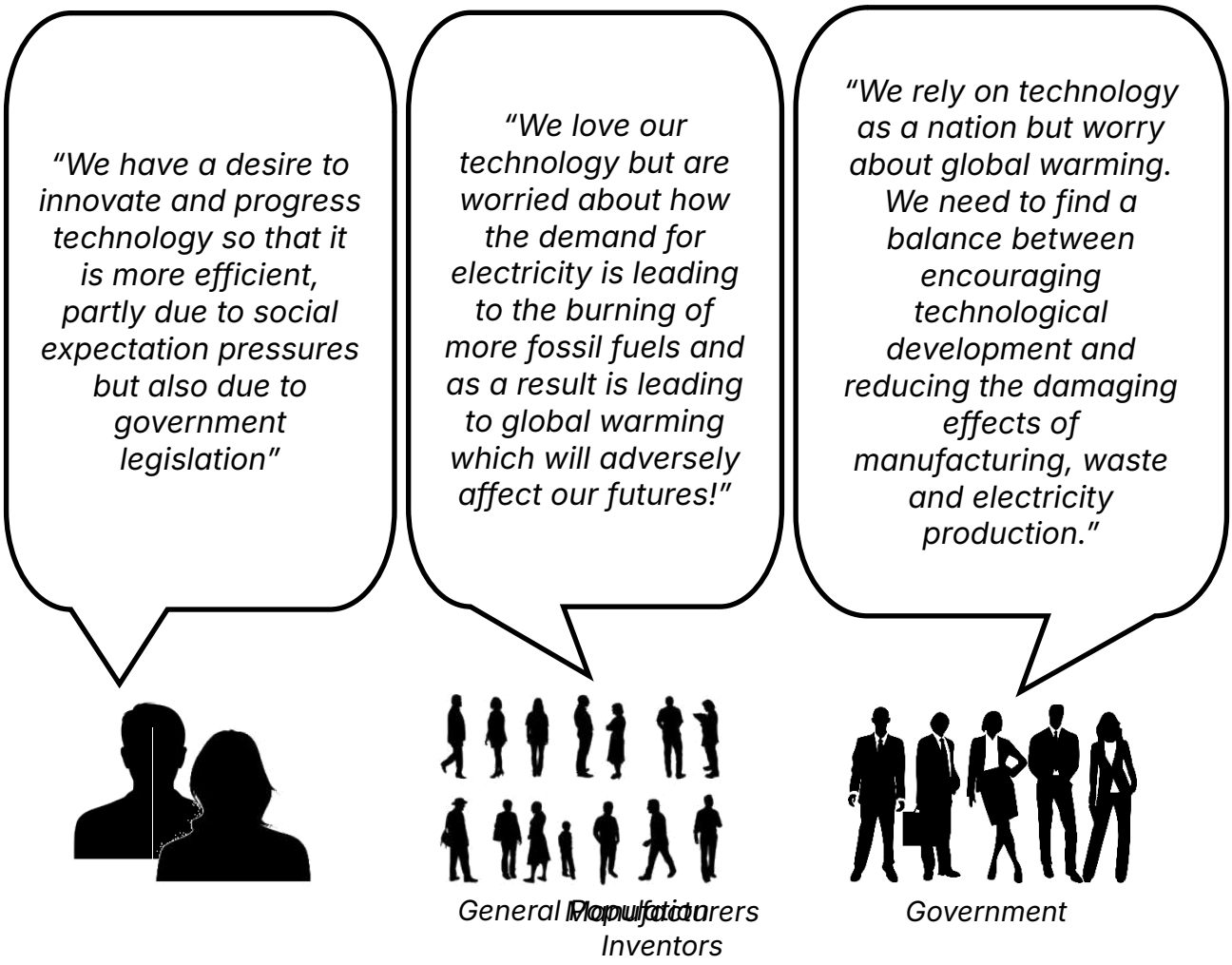
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harm. This is especially true with mobile devices which contain batteries full of harmful chemicals.

Due to ethical and legal expectations, there have been a number of changes which have led to the development of greener technologies which are more efficient and therefore require less electricity.

Social expectations and legislation have led to progress in how waste is dealt with – recycling is now the norm, for example.

Stakeholder's Views



Erosion of Privacy

In today's society, humans are using social media and are forced to provide personal information to government systems, doctors, the police etc.

This is often for good reasons, for example, if you went to the doctors, you would expect them to know your medical history.

However, when you put all these systems together, a fairly complete picture about each individual can be formed.

Ethically is it right that your privacy could be exposed in this way?

Monitoring humans so that those who pose threats to society can be identified can be a good thing.

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However, if all this information gets into the wrong hands it can have terrible consequences – for example, identity theft can cause individuals to lose their savings, homes and be accused of crimes that they didn't commit.

In addition to this, advertising companies can now easily collect information about individuals (often without their knowledge) and then target them with adverts. This can be annoying.

Stakeholder's Views

"Our privacy is no longer private, our complete identity can be pieced together though online services. On a positive note, this knowledge can keep us safe as threats can be better monitored"



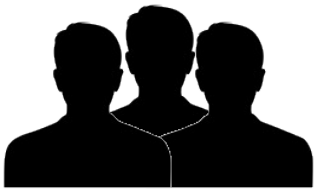
General Population

"There is a huge amount of pressure on us to protect our citizens and this is often done through intelligence and surveillance. However, there is social unrest with the way in which our methods are eroding the privacy of our citizens. Do we trust our citizens or do we monitor in order to keep them safe?!"



Government

"Greater monitoring makes it harder for us to carryout our illegal operations and is sending us more 'underground', forcing us to develop new ways to avoid surveillance. They can't stop us – they can just make it harder for us!"



Criminals

The Digital Divide

There are parts of the UK with super fast broadband and other parts which still have slow or no broadband at all.

This causes issues. Now-a-days, productivity is linked with communication speed – you can get more work done and thus attain greater profits with faster broadband speeds.

If half the country has slower broadband, this can cause unfair advantages to certain organisations over others.

This digital divide is seen between countries too.

For those countries with low IT literacy levels and skills, they are less likely to be able to compete in the modern world – thus worsening the divide further. This clearly is a cultural issue, with some cultures who may not be so technology immersive, suffering over time.

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Stakeholder's Views

"Our businesses are thriving due to faster communication speeds. We can get more work done, more quickly, due to our faster internet speeds."
"Due to our education, using a computer is second nature and as a result we have greater prospects in life."
"Due to our country being technologically advanced, our health care, security, transportation etc. are all advanced leading to a happy way of life."



The Advantaged

"Our internet speeds are slow and as a result we are not as productive as other similar companies. We are struggling to compete with them."
"Our education system has little money and our school has only one computer. It is difficult for all students to develop their ICT skills which may stifle our job prospects in life."
"Our country is poor and cannot invest in technology like other countries. Many of our services are quite limited and this affects our daily life adversely."



The Disadvantaged

Advantaged countries will use technology to further develop (medical research, transportation, education etc.) further widening the gap.

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Mobile Technology

The last several years has seen advancements in mobile technology. This has led to a range of advantages and disadvantages to employers and employees.

With mobile technology, workers can be more productive. They no longer need to be 'at the office' to work. They can work whilst on the move or when at home.

This is great for employers as their workforce can be more productive. Employees may also feel the benefits by having more flexibility with their working hours.

A major negative issue with this is that workers may struggle to "switch off" from work, which can adversely affect their home/work life balance.

Also, mobile devices are much easier to lose or be stolen and if there is sensitive information on the devices, this can be a major problem for organisations – the company may be in breach of the 'Data Protection Act'.

Furthermore, being connected 24/7 can lead to addiction too which again can adversely affect lives.

Stakeholder's Views

"I like being connected with mobile technology as it gives me freedom to choose my own working hours. This actually reduces my stress levels".
"I struggle with the fact that I have mobile technology as I find that I cannot ever switch off from work and from the digital world. This leads me to struggle with my home/work life balance - I sometimes find myself checking my work emails in the middle of the night!"



Workers

"We love it now that our workers have mobile technology. Because they are always connected we have seen productivity rates increase which has led to a rise in profits."
"We are having to support our workers a lot more now with stress."
"We are also dealing with more instances of sensitive company information getting into the wrong hands as people are storing company data on their mobile devices and then misplacing their devices without proper security – we hope it doesn't ever fall into the wrong hands!"



Companies

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Social Media

Recent years has seen the rise in Social Media. This has created a number of issues.

On a positive note, people are much more connected these days and as a result are able to share information and ideas with ease.

It has also led to greater democracy in some countries where state secrets have been exposed leading to societies demanding better treatment and more of a say in how countries are run.

Social media has also led to some negative issues such as trolling, cyber bullying and other forms of online abuse.

Stakeholder's Views

"I love social media. I can connect to my friends and share knowledge and ideas with them."
"I have been able to learn some hidden truths about how my country is run which has led to me fighting for change in how our country is run".
"Social media has ruined my life. Cyberbullying and Trolling has made my life unbearable!"



Society

"Social media allows us to check up on our workers and see if a prospective worker is right for our job!"
"Social media has caused issues for our company as one person decided to post about a negative experience they had with us – it went viral and we are now trying to rebuild trust with our customers".



Companies

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Health Issues

With a greater dependency on computers, people are spending far more time at a computer, which, as a result, means that in their daily lives they are spending less time being active.

This has led to a rise in the number of people being obese, which in turn greatly increases the number and severity of health problems for those individuals. This also adds pressure on the NHS.

Another health issues from repeated use of a computer is backache and repetitive strain injury (RSI).

Also, addiction can result from over use of computers, for example online betting and gaming. Addiction can also adversely affect lives in a number of ways.

Stakeholder's Views

"Due to the demands from my job I am forced to work at a computer for much of the day and this is leading to health problems – I have terrible backache, RSI and I am putting on weight".
"I find that I cannot switch off from the digital world – I am addicted to my mobile phone which is causing me to being detached from the real world – I don't eat or sleep properly anymore and I have fallen out with my friends.".



Society

"We are spending far more of our budget dealing with health problems relating to obesity. I wish people looked after themselves better then we would be under far less pressure as a health service.".
"We (the government) need to educate society on healthy living to ensure that our society's health improves, reducing the strain on the health service."



Government . NHS

Changing our Cultures

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Computer technology has been responsible for a change in our culture over the last few years. 20 years ago, the terms 'Selfies', 'Viral Videos' and 'Blogging' were pretty much unheard of but now they are part of our language and (to some degree) part of our behaviour.

Society seems to be forever trying to get the perfect 'Selfie' – we even have an invention to help us take them – 'the selfie stick!'. We seem to be forever filming in the hope that we will capture a moment that will make us famous. We feel the need to document all that we do, day in day out, in a blog.

Some real positives have come from this. The fact that we are forever filming has led to the capture of criminals, who had been unwittingly filmed 'in the act' and the filming of major incidents, which we have been able to analyse and learn from.

Blogging has led to a greater understanding of people's differences and the sharing of information and ideas.

On a more negative side, constant filming as led to the erosion of people's privacy and to nasty crazes such as happy slapping.

Stakeholder's Views

"I am forever filming every moment in the hope that I capture something that will go viral and make me famous".
"The other day I was filming an air show and something went wrong. The plane crashed! – everyone was ok thank goodness. The police took my video and it helped them find out exactly what happened".
"I had a great idea the other day and so decided to start a blog! I now have 500 followers and they have all thanked me for sharing my ideas with them – I feel like I am making a difference to their lives!"



Society

Legislation (The Law)

As computers develop and computer use increases, people are finding more and more ways to use computers to their advantages but often to the disadvantage of others. More and more laws are being passed concerning computer use and this is due to the rapid way in which this technology develops.

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There are 5 major laws concerning computers which you should be aware of:

- Data Protection Act (1998)
- Computer Misuse Act (1990)
- Copyright, Design and Patents Act (1988)
- Creative Commons Licencing
- Freedom of Information Act (2000)

Data Protection Act (1998)

With computers now holding so much personal information about people, companies have to look after this data properly. The law forces companies to have highly secure systems and to only hold onto personal information that a company really needs. In order to comply with this law, organisations must first contact the government to say that they will be storing people's personal information. They must make it clear WHAT information they will be storing and how they will use it.

There are in fact 8 principles of the data protection act that organisations must follow:

1. Data must be used in a fair way
2. Data must only be used for the role it is intended for
3. Data must be relevant – cannot gather more than you need
4. Data must be up-to-date
5. Data must not be kept for longer than it is needed for
6. The person whom the data is about must able to access the data if they wish
7. Data must be kept safely and secure
8. Data must not be transferred without protection (i.e. encryption)

Computer Misuse Act (1990)

With the internet connecting people, businesses and the world, people can access sensitive data far more easily. The Computer Misuse Act aims to prevent sensitive data getting into the wrong hands and being edited or destroyed. The act has three main laws.

You must not:

- Gain unauthorised access to a network
- Modify data on a network without permission
- Create and / or supply Malware.

Copyright, Design and Patents Act (1988)

The World Wide Web contains vast quantities of ideas and information in a digital format. It is therefore very easy to copy and share other people's work, ideas and files. This law makes it illegal to copy and / or share other people's work and ideas (intellectual property) without acknowledgement and permission. The law doesn't just govern digital work – it also covers written work and people's ideas too! Although, when it comes to illegal downloading, it is becoming very difficult to enforce the law due to the number of people who download media without permission. Having said that, a number of torrent hosting sites have been prosecuted and shut down – but then they just host the site again in another country...!?

Creative Commons Licencing

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There are times that the owner of some intellectual property would like other people to be able to use, share and edit their work. To enable this to occur without being in breach of copyright law, the owner of some intellectual property can make their work available under the Creative Commons Licence. When they do this they can choose exactly how people can use their work:

- Attribution – Work can be copied, modified and shared but the owner must be acknowledged
- Share-alike – If work is modified it can only be shared with the same licence that the original piece of work had
- Non-Commercial – Cannot re-use the work for own profit
- No Derivative Works – Can copy and share but cannot modify

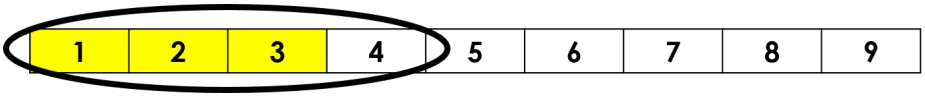
They can promote their work under one or a mixture of these licences.

Freedom of Information Act (2000)

This act allows the public to gain access to 'public' information held by public sector organisations (organisations that are funded by the tax payer). For example, the government, schools, police and the NHS must all publish certain information on a regular basis. This is often in the interest of transparency. If you were paying thousands of pounds in tax each year, you would probably like to know what your money has been spent on. The general public can also request certain information, such as the contact details of all schools in England. Information which may cause harm to national security can of course be held back from public view.

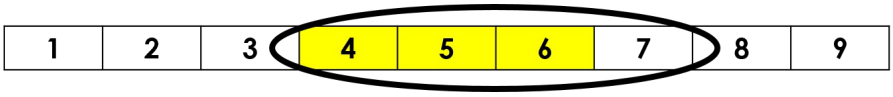
Questions *(The question zone you choose must either match your target grade or be higher!)*

Question Zone 1-3

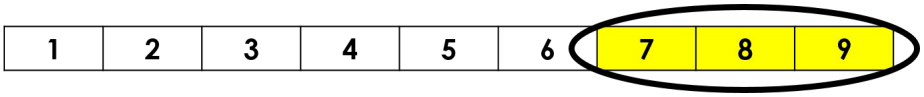


1. Describe some of the impacts that technology has on the environment and what has been done to try to reduce these impacts. [5]
2. What does the term 'Erosion of Privacy' mean? Your answer should include examples of how computer technology has led to the erosion of privacy. [5]
3. List the 5 major laws concerning computer technology and for each briefly describe their purpose. [5]

Question Zone 4-6



1. List the 5 major laws concerning computer technology and for each briefly describe their purpose. [5]
2. Discuss the effects that 'Social Media' has had on society. [8]
3. Discuss the effects that 'Mobile Technology' has had on society. [8]



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Question Zone 7-9

1. Discuss the effects that 'Mobile Technology' has had on society. [8]
2. Discuss how modern technology has influenced cultures. [8]
3. Discuss the effects that technology is having on our health. [8]

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples:** Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**.

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Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Ethical: *Discussing what is morally right and morally wrong.*
Legal: Discussing what is right and wrong in the eyes of the law.
Cultural: *Discussing how ethnic groups, countries, religions affect and are affected by technology.*

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State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
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Stick answer sheet here

Reflections ***Score:*** / ***Percentage:*** % ***Grade:*** ***Progress*** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

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Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts: